DRUGS · LOCAL ANAESTHETICS AND VASOCONSTRICTORS

Bupivacaine and levobupivacaine

Bupivacaine and its S-enantiomer levobupivacaine are long-acting amide local anaesthetics. They provide 4 to 8 hours of analgesia after wound infiltration, tonsillar fossa infiltration and nerve blocks. They are more cardiotoxic than lidocaine: toxicity can cause refractory ventricular arrhythmias and cardiac arrest, so the dose limit is lower.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

BUPIVACAINE AND LEVOBUPIVACAINE

	SUMMARY
Class	Long-acting amide local anaesthetics
How they work	Sodium-channel blockade; high lipid solubility and protein binding give long duration; slow dissociation from cardiac sodium channels causes cardiotoxicity
Preparations	0.25% and 0.5% (bupivacaine also with adrenaline 1:200,000); levobupivacaine 0.25%, 0.5%, 0.75%
ENT uses	Wound infiltration after neck, thyroid, parotid and ear surgery; nerve blocks (superficial cervical plexus for thyroid and neck surgery); post-tonsillectomy infiltration (variable evidence); scalp blocks
Maximum dose	Bupivacaine 2 mg/kg (up to 150 mg in 4 hours), with or without adrenaline; levobupivacaine 150 mg as a single dose (400 mg in 24 hours)
Quick conversion	0.25% = 2.5 mg/ml; 0.5% = 5 mg/ml
Onset and duration	Onset 10 to 20 minutes; duration 4 to 8 hours

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Infiltration and blocks

Infiltrating the wound at the end of surgery reduces early pain and opioid use. A **superficial cervical plexus block** at the midpoint of the posterior border of the sternocleidomastoid reduces pain after thyroidectomy and neck dissection. Mixing with lidocaine combines fast onset with long duration, but the toxic doses are additive.

2.2 Tonsillectomy

Infiltrating the tonsillar fossae may reduce early pain; rare complications (airway obstruction from parapharyngeal spread, cranial nerve palsies, and vessel injection) mean it is used selectively.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Cardiotoxicity (ventricular tachycardia and fibrillation, prolonged and difficult resuscitation) and neurotoxicity; levobupivacaine is slightly less cardiotoxic. Bupivacaine must never be used for intravenous regional anaesthesia. Manage toxicity as for lidocaine, with early **lipid emulsion**.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Maximum doses and why lower than lidocaine.
- Why bupivacaine toxicity is so dangerous to the heart.
- The superficial cervical plexus block.

4.2 Common pitfalls

COMMON PITFALLS

- Treating lidocaine and bupivacaine doses as separate when combined.

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